FLEMING

Course Outline

Course Title:	Geovisualization II		
Course Number:	GEOM106	**Approval Date:**	2022/1/9
Course Hours:	60 hours	**Academic Year:**	2021
Academic School:	School of Environmental& Natural Resource Sciences		
Faculty:	Marikka Williams - Marikka.Williams@flemingcollege.ca		
Program Co-ordinator or Equivalent:	Shawn Morgan - Shawn.Morgan@flemingcollege.ca		
Dean (or Chair):	Lisa Kraemer - lisa.kraemer@flemingcollege.ca		

Course Description

The discipline of cartography has long been about the art and science of communicating geospatial information through the incorporation of graphical symbology. Over the past two decades, geographic visualization, also referred to as geovisualization, has emerged to include not only the cartographic domain but also the understanding of the human cognition of information visualization and the inclusion of technology. As within numerous other scientific fields, the use of sophisticated interactive computer technologies has enabled geographers to render greater supporting geospatial data analysis through the use of interactive visualization. Collectively these factors are allowing for more advanced human-map interaction in revealing unknowns through visual communications.

Prerequisites: None.

Corequisites: None.

Work Integrated Learning: Service Learning

Experiential Learning: Interactive Simulations

Course Delivery Type

Online Synchronous.

All course hours delivered are online synchronously with specified meeting times. Delivery times are specified on the academic timetable.

Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Conceptualize and describe the importance of map design as an efficient and accurate method of information transfer between the map author and the map user.
2. Combine logical and semiotic perspectives to convey how maps represent reality with an emphasis on visual-cognitive processes.
3. Plan and manage projects, and develop workflows to accomplish specific objectives and tasks.
4. Describe and apply fundamental techniques used to retain a high degree of positional and attribute accuracy while capturing, structuring and displaying geographic data.
5. Acquire, explore, analyze, interpret, and integrate data from various sources (in different formats and in different coordinate systems) to explore and analyze spatial phenomena, develop visualizations and validate results.
6. Proficiently employ standard software packages used in GIS industries to acquire, prepare, analyze and display geographic information for a wide audience.
7. Create cartographic products utilizing point, line and area mapping techniques which are based on recognized data requirements and design principles that communicate geographic information effectively.
8. Synthesize the various stages involved in the abstraction of reality into an organized whole to facilitate cartographic communication.

Software requirements:

Currently this course focuses on the cartographic concepts, structures, and principles that are presented and not on the software used to complete assignments. ArcGIS Pro, ArcGIS for Adobe Creative Cloud, Adobe Illustrator and Avenza MAPublisher will be used for this course. Students taking this course will need keyboarding and basic computer skills such as file management. Graphic software experience will be an asset. Other commonly used software are Word for word processing, Excel for spreadsheets, and Access for data base. Basic introductory instructions will be given for each new software that is introduced. Selected user manuals may be recommended for specific software in conjunction with the teachings of this course.

Learning Resources

Required Textbook:

Field, Kenneth (2018) Cartography, ESRI Press, pp 576.

(Optional) Recommended Learning Resources:

Bertin, Jacques, Translated by William J. Berg (1983) Semiology of Graphics Esri Press, pp 460.

Slocum, Terry A., Robert B. McMaster, Fritz C. Kessler, and Hugh H. Howard. (2009). Thematic Cartography and Geographic Visualization 3rd ed. New Jersey, Pearson Prentice Hall Inc., pp 561.

Brewer, Cynthia A. (2015) Designing Better Maps: A Sourcebook for GIS Users 2nd ed. Esri Press, pp 260.

Bell, Sarah (2021) Mapping by Design: A Guide to ArcGIS Maps for Adobe Creative Cloud. , Available from: VitalSource Bookshelf, (1st Edition) Esri Press. pp. 225.

Peterson, Gretchen N. (2021) GIS Cartography: A Guide to Effective Map Design, Available from: VitalSource Bookshelf, (3rd Edition). Taylor & Francis, pp. 318.

Course notes, articles and website readings

Assessment Summary

Assessment Task	Percentage
In-class activities	15%
Assignments	25%
Applied Learning	45%
Tests	15%

Student Success: Policies and Procedures

Mutually, faculty and learners will support and adhere to college Academic Regulations, and Student Rights and Responsibilities. The following policies and guidelines have been developed to support the learning process.

Please click on the link for information about:

- Academic Integrity (2-201A) (https://department.flemingcollege.ca/hr/attachment/7750/download)
- Accessibility for Persons with Disabilities (3-341) (https://department.flemingcollege.ca/hr/attachment/5619/download)
- Grading and Academic Standing (2-201C) (https://department.flemingcollege.ca/hr/attachment/7752/download)
- Guidelines for Professional Practice: Students and Faculty (https://flemingcollege.ca/PDF/guidelines-for-professional-practice-students-faculty.pdf)
- Student Rights and Responsibilities (5-506) (https://department.flemingcollege.ca/hr/attachment/269/download)

If you will need academic accommodations (for example if you have a learning disability, mental health condition such as anxiety or depression or if you had an IEP in high school), please contact the Accessible Education Services (AES) department (https://department.flemingcollege.ca/aes/) to meet with a counsellor.

Alternate accessible formats of learning resources and materials will be provided, on request.

Program Standards

The **Ministry of Colleges and Universities** oversees the development and the review of standards for programs of instruction. The **Ministry of Labour Training and Skills Development** oversees the development and the review of standards for programs of instruction for Apprenticeship training in the province of Ontario. Each college is required to ensure that its programs and program delivery are consistent with these standards, and must assist students to achieve these essential outcomes.

This course contributes to Program Standards as defined by the Ministry of Colleges and Universities (MCU). Program standards apply to all similar programs of instruction offered by colleges across the province. Each program standard for a postsecondary program includes the following elements:

- **Vocational standards** (the vocationally specific learning outcomes which apply to the program of instruction in question);
- **Essential employability skills** (the essential employability skills learning outcomes which apply to all programs of instruction); and
- **General education requirement** (the requirement for general education in postsecondary programs of instruction that contribute to the development of citizens who are conscious of the diversity, complexity and richness of the human experience; and, the society in which they live and work).

Collectively, these elements outline the essential skills and knowledge that a student must reliably demonstrate in order to graduate from the program. For further information on the standards for your program, follow the MCU link (www.tcu.gov.on.ca/pepg/audiences/colleges/progstan/).

Detail Plan

Term:	2022 Winter
Faculty:	Marikka Williams - Marikka.Williams@flemingcollege.ca
Program Co-ordinator or Equivalent:	Shawn Morgan - Shawn.Morgan@flemingcollege.ca
Dean (or Chair):	Lisa Kraemer - lisa.kraemer@flemingcollege.ca

Learning Plan

Wks/Hrs Units	Topics, Resources, Learning, Activities	Learning Outcomes	Assessment

Wks/Hrs Units	Topics, Resources, Learning, Activities	Learning Outcomes	Assessment
Week 1	Lecture : Course Introduction; Maps as Information Products; Lab : Technical Focus Exercise (All) - Map Design Specifications and Time Estimations	1-3	
Week 2	Lecture : Revisiting Cartographic Design: Setting the Stage; Lab : Technical Focus Exercise (All) - Acquiring Data and Building Balanced Map Layouts	1-6	
Week 3	Lecture : Statistical Foundations, Data Classification and Quantitative Methods; Lab : Technical Focus Exercise (All) - Exploring Data with Graphs and Stats	3-6	
Week 4	Lecture : Choropleth Mapping Techniques (Lecture Focus Questions I Due); Lab : Technical Focus Exercise 1 - Area Mapping Techniques: Evaluating Algorithms and Color Palettes	1-7	Lecture Focus Questions I Due (5%)
Week 5	Lecture : Dasymetric Mapping Techniques; Lab : Technical Focus Exercise 1 - Area Mapping Techniques: Adding Context and Addressing Ecological Fallacy	1-7	
Week 6	Lecture : Isarithm Mapping Techniques; Lab : Technical Focus Exercise 1 - Area Mapping Techniques: Interpolating Information	1-7	
Week 7	Lecture : Cartogram Mapping Techniques; Lab : Technical Focus Exercise 1 - Area Mapping Techniques: Accounting for Area or Amount	1-7	
Week 8 Independent Learning Week	Program Oriented, self-directed studies/field activities		
Week 9	Lecture : Proportional Dot Mapping Techniques; (Lecture Focus Questions II Due); Lab : Technical Focus Exercise 2 - Proportional Dot Mapping Techniques	1-7	Lecture Focus Questions II Due (5%) Technical Focus Exercise 1 (15%) Due (prior to lab session)
Week 10	Lecture : Dot Density Mapping Techniques; Lab : Technical Focus Exercise 2 - Dot Density Mapping Techniques	1-7	Final Project Milestone I Due (5%)
Week 11	Lecture : Line Mapping Techniques; Lab : Technical Focus Exercise 3 - Qualitative Line Mapping Techniques (Technical Focus 2 Due)	1-7	Lecture Focus Questions II Due (5%); Technical Focus Exercise 2 (15%) Due (prior to lab session)

Wks/Hrs Units	Topics, Resources, Learning, Activities	Learning Outcomes	Assessment
Week 12	Lecture : Visualizing Uncertainty; Lab : Technical Focus Exercise 3 - Quantitative Line Mapping Techniques	1-7	Final Project Milestone II Due (5%)
Week 13	Lecture : Visualizing Depth; Lab : Basic Terrain Visualization Mapping Techniques (Bonus Opportunity); Final Project Work Period (Technical Focus 3 Due)	1-7	Technical Focus Exercise 3 (15%) Due (prior to lab session)
Week 14	Lecture : Reviewing Cartographic Design (Lecture Focus Questions III Due); Lab : Final Project Progress Milestone III	1-8	Lecture Focus Questions III Due (5%) Final Project Milestone III Due (5%)
Week 15	Lecture : Final Exam;.Lab : Final Project Map Gallery	1-8	Final Exam (15%); Final Project Due (10%)

Assessment Requirements

Assessment Task	Date/Weeks	Course Learning Outcome	Percentage
Technical Focus Exercise X 3	Week 9, 11, 13	3-7	45%
Lecture Focus Question X 3	Week 4, 9, 14	1-4	15%
Final Project Milestones X 3	Week 10, 12, 14	3-8	15%
Final Exam	Week 15	1-4	15%
Final Project	Week 15	3-8	10%

Students are responsible for submitting assessed work as instructed on the due date and time indicated in writing, unless written arrangements have been made. Online submission time will determine late submissions. If the assessment is not submitted by the due date and time, a penalty of 10% of the assessment value per day will be assigned starting after the due date and time. This penalty applies for each 24-hour period, including weekends. After 120 hours (5 days) submissions will not be accepted.

An individual extension of the due date may be granted at the discretion of the course faculty if the student has submitted a written request. It is the student's responsibility to notify the course faculty of any problems as soon as possible. Considerations for extensions include prescribed accommodations, medical certificate or legal document. Late submissions will not be accepted under any circumstances after the assessment has been graded and distributed to the class.

Final grades in this course are assigned based on the level of academic achievement according to the Learning Plan and Assessment Requirements cited in this course outline. Faculty will not offer additional evaluation activities to individual students beyond those cited in this course outline.

Appeals of any term work evaluation must be initiated within a 5 business day period after receiving results. Appeals, as with any other academic inquiry or concern, should first be first directed to the instructor, and then,

failing satisfactory resolution, as specified by the Academic Regulations Policy.

Turnitin Statement

Students agree that by taking this course all required papers may be subject to submission for textual similarity review to Turnitin.com in support of academic integrity. All submitted papers will be included as source documents in the Turnitin.com reference database solely for the purpose of detecting plagiarism of such papers. Use of the Turnitin.com service is subject to the Usage Policy posted on the Turnitin.com site.

If students choose to opt out of using the software, they must inform their faculty member at the time of the assignment, of their refusal and meet with the faculty to discuss their options. For further information on the use of Turnitin, please refer to the Operating Procedure 2-201F (https://department.flemingcollege.ca/hr/attachment/10233/download)

Exemption Contact

Marikka Williams

Office: Frost Campus 214

Email: Marikka.Williams@fleming.college.ca

You will also need to receive clearance from the Registrar's Office and the Admissions Office.

Prior Learning and Assessment and Recognition (PLAR)

PLAR uses tools to help learners reflect on, identify, articulate, and demonstrate past learning which has been acquired through study, work and other life experiences and which is not recognized through formal transfer of credit mechanisms. PLAR options include authentic assessment activities designed by faculty that may include challenge exams, portfolio presentations, interviews, and written assignments. Learners may also be encouraged and supported to design an individual documentation package that would meet the learning requirements of the course. Any student who wishes to have any prior learning acquired through life and work experience assessed, so as to translate it into a college credit, may initiate the process by applying through the Registrar's office. For more information please click on the following link: http://flemingcollege.ca/admissions/prior-learning-assessment-and-recognition

Course Specific Policies and Procedures

It is the responsibility of the student to retain this course outline for future reference. Course outlines may be required to support applications for advanced standing and credit transfer to other educational institutions, portfolio development, PLAR and accreditation with professional associations.

Synchronous sessions may be recorded. As a result, your image, voice, name, personal views and opinions, and course work may be collected under legal authority of section 2 of the Ontario Colleges of Applied Arts and Technology Act, 2002. This information will be used for the purpose of supporting student learning. Any questions about this collection can be directed to the Privacy and Policy Officer at freedomofinformation@flemingcollege.ca or by mail to 599 Brealey Drive, Peterborough, ON K9J 7B1.

DEFINITIONS

Professor: The individual(s) who are employed by Fleming College and assigned to deliver and assess this course's approved learning outcomes in support of the program's vocational learning outcomes.

Student: Any person who is listed by the registrar as enrolled for this course while it is active.

Student Work: Any submitted assignment, exercise, presentation, product, report, code, map, idea, solution, database, web page, program, or model created by the student, in whole or in part, digital or physically printed, under the direction of the professor, where it fulfills one or more academic requirements of a class during the academic semester when the student is actively enrolled.

Group Work: Individual students may be required to work together with other students who are also enrolled in the same class.

Academic Integrity: As defined by the Academic Regulations Policy of Fleming College.

POLICIES AND PROCEDURES

Accommodation: Students who require accommodations under the Accessibility for Ontarians with Disabilities Act (AODA 2005) should ensure that their professor has received a copy of their accommodation plans.

Recording: Permission from the instructor must be obtained prior to making any live video or sound recordings of the class. Any approved recordings are to be used personally and cannot be shared and must be destroyed/deleted when the course ends.

Devices: All electronic devices (e.g. cell phones, laptops, etc.), except those approved by Accessible Education Services or professor, must be turned off or silenced for the duration of the class. Personal communication to outside entities is not permitted during the class including: Social Media (Facebook etc), Mobile Phones, Chat (SMS, Talk, Snapchat, Messenger, etc.) or any other communication unrelated to, or a distraction to the class.

Academic Integrity: Each student has the responsibility to support academic integrity. Students may be expected to work as individuals or in groups on assessments (as indicated on each assessment). No individual or group member should have direct input into another individual's or group's Student Work. Breaches of academic integrity (such as, but not limited to: plagiarism, fabrication, deception, cheating, bribery/paid services, sabotage, impersonation) will be addressed using the approved Academic Regulations Policy of Fleming College. Sharing of digital or physical copies of any Student Work for a marked assessment is considered a breach of academic integrity. Students must hand in original Student Work they (or their group) created from the current academic semester.

Lateness or Absence: Students who are late for class are a disruption and have a negative impact on the learning environment. For reasons relating to classroom management and/or laboratory/field safety, students arriving after class has commenced may be refused entry. Absences will be dealt with by the professor in accordance with Fleming College's Class Absence Operating Procedure.

Behaviour: Student behaviour in all learning environments (e.g. class, lab, field, on-line, placement, etc.) must abide by Fleming College's Student Rights and Responsibilities Policy.

Computing Resources and Software: At no time are students to contact vendors for assistance for provided software or services. All computing systems used in any part of the course are subject to the same terms and conditions under the Information Technology Appropriate Use Policy.

Public Sharing of Projects and Assignments: Any submitted Student Work may be made public for the promotion of the program and/or provided as positive examples to future enrolled students. If an individual does not wish to have their Student Work shared, it is required they submit a written request with explicit instructions on what is to be excluded to each instructor for each course.

Final Grades: Final grades in this course are assigned based on the level of academic achievement according to the Learning Plan and Assessment Requirements cited in this course outline. Faculty is not obligated offer additional evaluation activities to individual students beyond those cited in this course outline.

Modifications to the Learning Sequence in this outline may occur to facilitate the learning of the students. Notification of any changes will be posted on D2L Brightspace. Additional time will be required outside of class for completion of learning activities.